

Problem Statement: CYBERGUARD - AI Powered Cyber Threat, Phishing & Digital Impersonation Detection and Response System

Domain: Cybersecurity + Artificial Intelligence

Background

The rapid adoption of digital platforms, online communication, cloud services, social media, digital payments and e-governance systems has significantly increased exposure to sophisticated cyber threats.

Attackers are increasingly using Artificial Intelligence and Generative AI to create highly convincing phishing emails, fake websites, malicious links, cloned voices, deepfake videos, fraudulent identities and personalised social-engineering attacks.

At the same time, organisations face threats such as abnormal login attempts, credential theft, account takeover, malware, suspicious network behaviour, API abuse and data-exfiltration attempts.

Traditional rule-based security systems are often inadequate to identify such evolving and AI-generated threats.

There is therefore a need for an AI-powered intelligent cyber-defence platform capable of detecting technical cyber threats as well as human-targeted phishing, impersonation and deepfake-based attacks.

Challenge

Design and develop an AI-powered Cyber Threat, Phishing and Digital Impersonation Detection & Response System capable of identifying, analysing, explaining and responding to emerging cyber threats in near real time.

The proposed system should use technologies such as Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, behavioural analytics, anomaly detection or Generative AI to analyse multiple sources of digital activity and identify suspicious or malicious behaviour.

The system should not merely generate alerts. It should also provide:

- Threat classification
- Risk/severity score
- Explanation of the detected threat

- Evidence or indicators contributing to the decision
- Recommended preventive or response actions

Key Threat Scenarios

Teams may address a combination of the following:

1. AI-Powered Phishing Detection

Detect suspicious:

- Emails
- SMS/messages
- Social-media messages
- QR-code-based phishing
- Fraudulent websites
- Malicious or deceptive URLs

The system should identify indicators such as urgency, impersonation, suspicious domains, malicious links, credential requests and unusual communication patterns.

2. Deepfake Detection

Identify or assess potentially manipulated:

- Images
- Videos
- Voice/audio clips
- Video calls
- AI-generated multimedia content

The system may generate an authenticity/confidence score and highlight indicators of possible manipulation.

3. Digital Impersonation Detection

Detect attempts to impersonate:

- Government officials
- Senior management
- Teachers or university authorities
- Financial institutions
- Organisations or brands
- Friends, relatives or known contacts

The system may analyse identity, communication style, metadata and behavioural patterns.

4. Credential Theft & Account Takeover Detection

Identify abnormal behaviour such as:

- Multiple failed login attempts
- Password spraying
- Login from unusual locations
- New or unknown devices
- Suspicious session activity
- Sudden changes in account behaviour

5. Malicious URL & Website Detection

Analyse URLs and websites for indicators including:

- Domain spoofing
- Look-alike domains
- URL manipulation
- Malicious redirects
- Suspicious SSL/domain characteristics
- Fake login pages

6. Intelligent Cyber Threat Detection

The solution may additionally detect:

- Malware indicators
- Suspicious network traffic
- API abuse
- Data-exfiltration behaviour
- Abnormal user activities
- Insider threats
- Unusual system or application logs

Expected Solution

The proposed prototype should preferably contain the following modules:

A. Multi-Source Threat Analysis Engine

The system may analyse:

- Emails
- SMS/messages
- URLs
- Images
- Audio
- Videos
- Authentication logs
- System logs
- Network traffic
- API logs

B. AI-Based Detection Engine

Participants may use:

- Machine Learning
- Deep Learning
- Natural Language Processing
- Computer Vision
- Speech/Audio Analysis
- Large Language Models
- Behavioural Analytics
- Anomaly Detection
- Graph Analytics
- Hybrid Rule + AI Models

C. Threat Risk Scoring

Every suspicious event should be assigned an appropriate risk level, for example:

Safe → Low → Medium → High → Critical

The system should explain the major factors behind the score.

D. Explainable AI

Instead of providing only:

“Phishing Detected”

the system should provide an explanation such as:

“High Risk: The sender domain closely resembles an authorised organisation, the message requests urgent credential verification, and the embedded URL redirects to an unrelated domain.”

For deepfake detection, the system may similarly explain the indicators influencing its confidence score.

E. Intelligent Response Recommendation

Depending on the detected threat, the system may recommend:

- Block suspicious URL
- Quarantine email
- Warn the user
- Require additional authentication
- Revoke active session
- Block suspicious IP/device
- Flag multimedia for manual verification
- Report impersonation
- Notify administrator/SOC
- Escalate the incident for investigation

F. Cybersecurity Command Dashboard

Develop an interactive dashboard displaying:

- Total events analysed
- Threats detected
- Threat category
- Risk level
- Phishing attempts
- Impersonation attempts
- Suspected deepfakes
- Account takeover attempts
- Attack timeline
- Frequently targeted users/services
- Recommended actions
- Incident status

Innovation Opportunities

Participants are encouraged to explore advanced capabilities such as:

- Generative AI-assisted cybersecurity
- AI-generated phishing detection
- Multimodal deepfake detection
- Voice-cloning detection
- Email sender authenticity analysis
- Digital identity verification
- Behaviour-based fraud detection
- Explainable AI
- Graph-based cyberattack analysis
- Real-time threat intelligence
- MITRE ATT&CK mapping
- Zero-day anomaly detection
- Privacy-preserving AI
- Federated learning
- Autonomous cyber-defence agents
- Automated incident-response playbooks

Minimum Expected Prototype

Each team should demonstrate at least **three different cybersecurity scenarios**, of which preferably:

1. One should involve **phishing/social engineering**.
2. One should involve **digital impersonation/deepfake or identity fraud**.
3. One should involve a **technical cybersecurity threat or abnormal behaviour**.

The prototype should demonstrate:

Detection → Classification → Risk Assessment → Explanation → Alert → Recommended Response

Minimum Deliverables

Teams should submit/demonstrate:

1. Working prototype.
2. Threat-detection mechanism.
3. Detection of at least three cybersecurity scenarios.
4. Risk-scoring mechanism.
5. Explainable threat assessment.
6. Cybersecurity monitoring dashboard.
7. Recommended mitigation/response mechanism.

8. System architecture.
9. Details of models/algorithms used.
10. Demonstration using simulated, authorised or publicly available cybersecurity datasets.
11. Accuracy/performance evaluation.
12. Scalability and deployment approach.

Desired Impact

The proposed solution should have the potential to evolve into an intelligent AI Cyber Defence & Digital Trust Platform that could be deployed for academic, government, and enterprise bodies. The objective is to create a system capable of protecting both the technology layer and the human layer of cybersecurity.